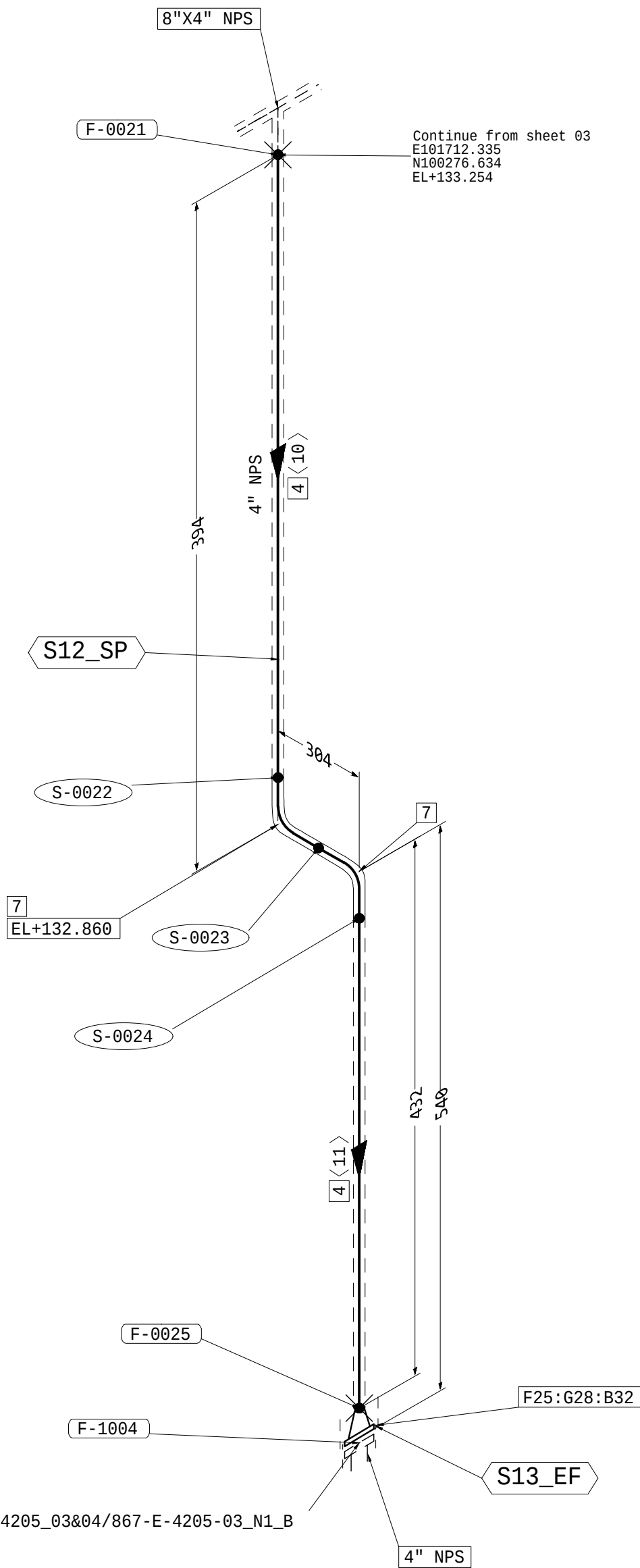
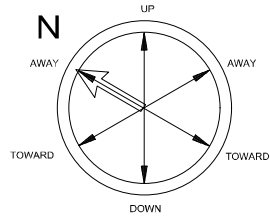


THIS SHEET MARKUP DRAWING IS GENERATED FROM ENGINEERING PCF & REFERRING APPROVED ENGNG ISOMETRICS. ENGINEERING ISOMETRICS TO BE REFERRED FOR REVISION APPROVAL INFORMATION & IN CASE OF ANY DISCREPANCY DATA GIVEN IN ENGINEERING ISOMETRIC SHALL GOVERN



CONT. TO: 867-E-4205\_03&04/867-E-4205-03\_N1\_B  
4 in RFFE 600  
E101712.335  
N100276.330  
EL+132.317

NOTES CONT. :

91. FOR SOUR SERVICE PWHT IS MANDATORY FOR 1. DOUBLE SIDED WELDING, 2. INTERNAL WELD REPAIR 3. INTERNAL WELD BUILD UP FOR DISSIMILAR THICKNESS JOINTS TO OVERCOME MISALIGNMENT.

| PWHT | CHEMICAL<br>CLEANING | PIPING<br>MATERIAL<br>CLASS | DESIGN            |                  | OPERATING        |                  | TEST<br>PRESS.<br>(PSIG) | TEST<br>MEDIUM | RADIOGRAPHY % |       | INSULATION   |      | TRACING | PAINTING | STRESS<br>SYS NO. | LINE ID         |
|------|----------------------|-----------------------------|-------------------|------------------|------------------|------------------|--------------------------|----------------|---------------|-------|--------------|------|---------|----------|-------------------|-----------------|
|      |                      |                             | TEMP.<br>(DEG. F) | PRESS.<br>(PSIG) | TEMP.<br>(DEG.F) | PRESS.<br>(PSIG) |                          |                | SHOP          | FIELD | THK.<br>(MM) | TYPE |         |          |                   |                 |
| Y    | NOTE-8               | 1CH2P06                     | 270               | 170              | 172              | 100              | 428                      | WATER          | 10            | 10    | 25           | HC   | NA      | APCS-2A  | 867-GPT4-154      | MDEA-17357-420A |

## MATERIALS

| PT<br>No.        | DESCRIPTION                                                                                      | NPD(IN) | ITEM CODE          | QTY   |
|------------------|--------------------------------------------------------------------------------------------------|---------|--------------------|-------|
| <b>PIPES</b>     |                                                                                                  |         |                    |       |
| 4                | PIPE SMLS 4" A106-B SOUR STD BE B36.10M                                                          | 4       | PPS2AERT04ZZABUAIA | 0.522 |
| <b>FITTINGS</b>  |                                                                                                  |         |                    |       |
| 7                | ELB 90° SMLS LR 4" A234-WPB SOUR STD BE B16.9                                                    | 4       | PE92AZRT04ZZABUAIQ | 2     |
| <b>FLANGES</b>   |                                                                                                  |         |                    |       |
| 25               | FLG WNR 4" A105N SOUR STD, 600# 125 B16.5                                                        | 4       | PFY2ADRT04ZZAEMADH | 1     |
| <b>GASKET</b>    |                                                                                                  |         |                    |       |
| 28               | GSK SPR WND W/CS CENTERING RING + I - SS 4" 4.50 MM THICK SS316L+GRPH FILL SOUR 600# B16.20      | 4       | PGUUB5RT04AIAADZCC | 1     |
| <b>FASTSNERS</b> |                                                                                                  |         |                    |       |
| 32               | BOLT, STUD+2 NUTS A193 GR B7M/ A194 GR 2HM, CERAMIC FLP COAT SOUR B18.2.1/18.2.2 - 160 mm Length | 0.875   | PH8ASCRTAH00160ZZF | 8     |

| PIPE CUT LENGTH SUMMARY |     |        |        |           |              | SXX_Spool Sequence Number (S01,02, SXX_SP - Fab. Spool<br>SXX_PP - Fab. P to P Doubling (0-6M+0M)<br>SXX_RP - Erec. Random Pipe (0M,12M)<br>SXX_LP - Erec. Part Loose Fittings<br>SXX_EV - Erec. Range (F-Loose Flange, FTS-Bolt, FTS-Spectacle)<br>SXX_EV - Erec. valve<br>SXX_EPG - Erec. Pipe Plug<br>SXX_SBF - Small Bore Spool Fab.<br>SXX_SBE - Small Bore Spool Erec.<br>SXX_SBD - Small Bore Spool Drain<br>SXX_SBV - Small Bore Spool Vent<br><br>[X] PART NUMBER<br><X> PIECE NUMBER<br>[SXX] SHOP WELD<br>[FSX] FIELD WELD<br>[SXX,01] SPOOL NUMBER |
|-------------------------|-----|--------|--------|-----------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SPL No.                 | NPD | PC No. | LENGTH | ALLOWANCE | TOTAL LENGTH |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| S12_SP                  | 4   | <10>   | 0.242  | 0.000     | 0.242        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| S12_SP                  | 4   | <11>   | 0.280  | 0.000     | 0.280        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         |     |        |        |           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         |     |        |        |           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         |     |        |        |           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         |     |        |        |           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         |     |        |        |           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         |     |        |        |           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         |     |        |        |           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

ISO For

## NOTES :

- ALL COORDINATES AND ELEVATIONS ARE IN 'METER' AND ALL DIMENSIONS ARE IN 'MM'.
- PIPE QUANTITY IS IN METERS AND OTHER QUANTITY IN NUMBERS.
- FOR STANDARD SUPPORT, REFER DOC. NO. "PP-868-ST-0001".
- IN CASE OF ANY DISCREPANCY FOR LINE CONDITIONS, LINE DATA GIVEN IN LINE LIST SHALL GOVERN
- FOR SMALL BORE LINES, DIMENSION & ROUTING SHALL BE FIELD ADJUSTED.
- THE SPECIFIED THEORETICAL LOCATION OF EQUIPMENT NOZZLES, SUPPORTING STRUCTURE SHALL BE VERIFIED BEFORE INSTALLING THE PIPING & INTOLERABLE DISCREPANCIES SHALL BE DISCUSSED WITH DESIGN SITE ENGINEER.
- ALL THREADED CONNECTION SHALL BE SEAL WELDED, EXCEPT DRAIN AND VENT PLUG.
- REFER LINE LIST.
- ALL PRESSURE WELDS, OTHER THAN BUTT WELDS, INCLUDING BRANCH CONNECTIONS SUCH AS BRANCH-OLETS, AND BOSS) SHALL BE 100% PT/MT EXAMINED. ALL ATTACHMENT WELDS ON PRESSURIZED COMPONENTS THAT ARE NOT HYDROTESTED AFTER MAKING THE ATTACHMENT WELD SHALL BE 100% PT/MT EXAMINED.

|                |                      |
|----------------|----------------------|
| BUTT WELD      | FIELD WELD           |
| INSULATED PIPE | TRACED PIPE          |
| PIPE SUPPORT   | PLATFORM PENETRATION |
| SOCKET WELD    | SCREWED JOINT        |

## STANDARD DETAIL REFERENCE DRAWINGS

DETAILS HEAVY WELDING BOSS FOR THREADED CONNECTIONS  
CONSTRUCTION DETAILS AND DETAILS FOR ISOMETRIC PIPING  
INSULATION DETAILS FOR HOT PIPING  
INSULATION DETAILS FOR COLD PIPING  
INSTALLATION OF JACK SCREWS FOR FLANGED JOINTS

AE-036175  
AD-036077  
AA-036911  
AA-036913  
AC-036930

## NOTES

- MATERIAL SHOWN WITH A BROKEN LINE AND SPOOLS LISTED IN THE SAUDI ARAMCO FABRICATION LIST ARE NOT TO BE FURNISHED BY THE VENDOR.
- PIPING DETAILS SHALL BE CUT AT ALL POINTS MARKED F.W. (FIELD WELD).
- EACH FABRICATED SECTION SHALL BE CLEARLY MARKED WITH ITS CODE NUMBER.
- UNLESS OTHERWISE NOTED: (A) ALL DIMENSIONS ARE TO FACE OF FLANGES, (B) BOLT HOLES ON FLANGES STRADDLE C. (C) WELDING ELBOWS ARE LONG RADIUS.
- FABRICATION SHALL BE IN ACCORDANCE WITH SAUDI ARAMCO SPECIFICATION 01-SAMSS-10.
- GASKET ALLOWANCE
- PIPING MATERIAL IS

| PIPE SIZE | SCHEDULE | SPECIFICATION |
|-----------|----------|---------------|
|           |          |               |
|           |          |               |
|           |          |               |
|           |          |               |

5. WELDING STRESS RELIEF - YES ☐ NO ☐ 100% X-RAY ☐ SPOT X-RAY 01-SAMSS-10 ☐
9. TESTING SHOP HYDROSTATIC TEST - YES ☐ NO ☐ THE SHOP HYDROSTATIC TEST SHALL BE CARRIED OUT ON PIECES HAVING ALL ENDS FLANGED, HYDROSTATIC TEST PRESSURE
10. ALL PIPING MATERIAL IS IN ACCORDANCE WITH SAUDI ARAMCO ENGINEERING SPECIFICATION SAES-L-105 CLASS . ALL EXCEPTIONS ARE NOTED ON THE ISOMETRIC.
11. INSULATION REQUIRED - YES ☐ NO ☐ INSULATION SHOWN ON DRAWING

## SPOOLS TO BE FABRICATED BY VENDOR

| NO.<br>PIECES | MARK NO. | SIZE | WEIGHT<br>LBS | NO.<br>PIECES | MARK NO. | SIZE | WEIGHT<br>LBS |
|---------------|----------|------|---------------|---------------|----------|------|---------------|
|               |          |      |               |               |          |      |               |
|               |          |      |               |               |          |      |               |
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|               |          |      |               |               |          |      |               |
|               |          |      |               |               |          |      |               |

## SPOOLS TO BE FABRICATED BY SAUDI ARAMCO

| NO.<br>PIECES | MARK NO. | SIZE | NO.<br>PIECES | MARK NO. | SIZE |
|---------------|----------|------|---------------|----------|------|
|               |          |      |               |          |      |
|               |          |      |               |          |      |
|               |          |      |               |          |      |
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|               |          |      |               |          |      |
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## REFERENCE DRAWINGS

|                           |                |
|---------------------------|----------------|
| PIPING PLAN               | YA-212287      |
| P&ID NO.                  | YA-221177-001  |
| PLOT PLAN KEY PLAN NO.    | YA-212083      |
| INDEX FOR PIPING ISO (AG) | PP-867-LI-0003 |
| INDEX FOR PIPING ISO (UG) | PP-867-LI-0004 |

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|-----|----------|-------------------------|----------|----------|----------|----------|
| 00A | 02-05-25 | ISSUED FOR CONSTRUCTION | AC/SN    | KY/PD    | BSP      | HJ       |
| REV | DATE     | DESCRIPTION             | PRO'D BY | CHK'D BY | APP'D BY | APP'D BY |

## REVISION STATUS

## REVISION VALIDATION

THIS REVISION NO. 00A IS COVERED FOR ALL APPROVAL AND CERTIFICATION REQUIREMENTS, PER SAEP-33A. UNDER DRAWING COMPLETION CERTIFICATE NO:

10-14815-0003-000

|                                         |                   |                                                                             |                 |         |
|-----------------------------------------|-------------------|-----------------------------------------------------------------------------|-----------------|---------|
| DESIGN CERTIFICATION<br>(DESIGN AGENCY) |                   | REVIEW FOR KEY DRAWINGS<br><input type="checkbox"/> REQUIRED (SAUDI ARAMCO) |                 |         |
| ENGG. DATE                              | CERTIFIED BY DATE | CONS. AGENCY DATE                                                           | OPERATIONS DATE | BY DATE |
| REV NO.                                 | DATE              | REVISION DESCRIPTION                                                        |                 |         |
|                                         |                   |                                                                             |                 |         |

JOB ORDER/REV NO. 10-14815-0003

THIS REVISION IS NOT TO BE USED FOR CONSTRUCTION UNLESS CERTIFIED AND DATED

SAUDI ARABIAN OIL COMPANY

ISOM 8" -MDEA-17357-1CH2P06

DRAWING TYPE  
ISO

MDEA SERVICE

JFGP GAS PROCESSING TRAIN-4

| JAFURAH   |       | SAUDI ARABIA   |          |          |
|-----------|-------|----------------|----------|----------|
| PLANT NO. | INDEX | DRAWING NUMBER | SHT. NO. | REV. NO. |
| 867       | L     | YB-215304      | 040f05   | 00A      |

SITE REV. NO. 00A

RESTRICTED

SMART

CADD-03

450420

09-05-2025

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